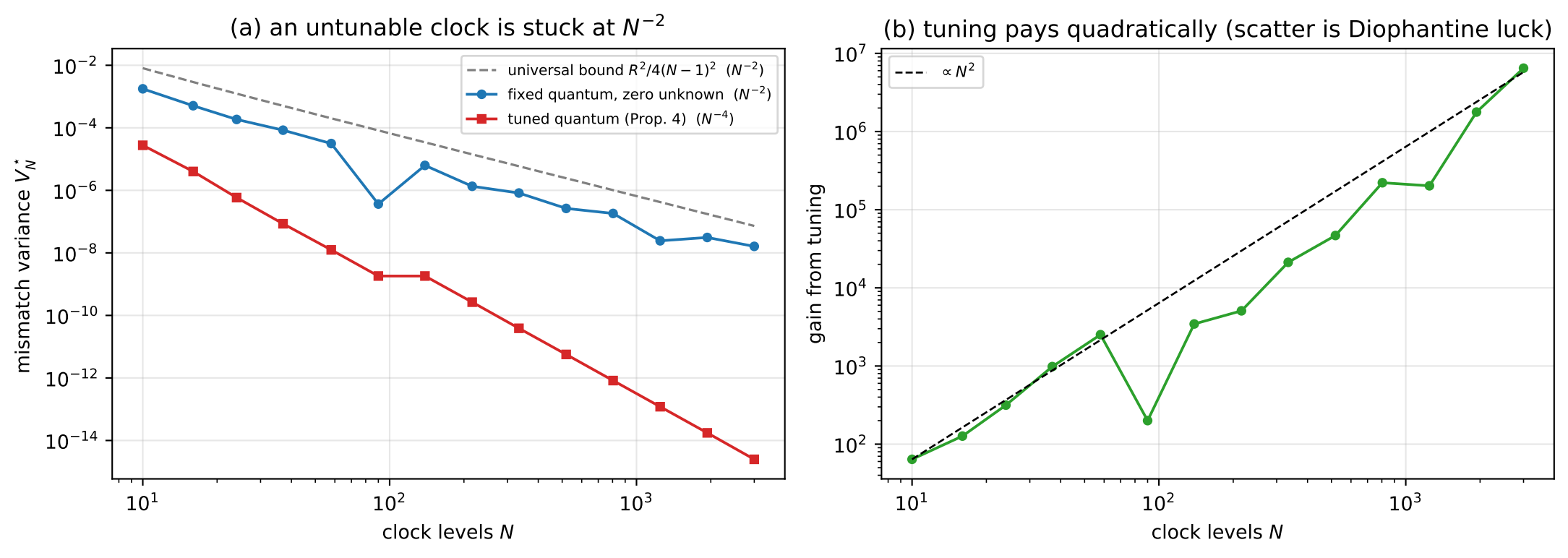
(a) an untunable clock is stuck at N^{-2}
universal bound $R^2/4(N-1)^2$ (N^{-2})
fixed quantum, zero unknown (N^{-2})
tuned quantum (Prop. 4) (N^{-4})
mismatch variance $V_N^\star$
clock levels N
(b) tuning pays quadratically (scatter is Diophantine luck)
$\propto N^2$
gain from tuning
clock levels N